

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – SHORT ANSWER TEST

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO5 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 3 – Short Answer Test
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.		
Student Name:	K. Pujitha	Reg. No.:	192524149
Section / Batch:	2 nd Year (AI & DS)	Date:	02/09/2026

Code of Conduct

I, K. Pujitha (192524149) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Pujitha

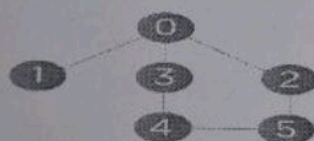
Instructions

- This test contains 10 short answer test questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:			100 Marks

Annexure A - Short Answer Test

- Perform the following operations in the Singly Linked List for the following input.
Input: 22-> 77-> 55-> 88->
 - Insert a new-node 110 at the end of the list.
 - Insert a new-node 101 at the position 4 of the list.
 - Delete a node at the beginning of the single linked list
 - Delete a node at the position 3 of the single linked list.
- Rotations in each step of insertion. 1,2,3,4,5,6,7,8,9,10 (), 4) Push (90), 5) Push (36), 6) Push (11), 7) Push (88), 8) Pop (), 9) Pop (). Draw the diagram of the stack and illustrate the above operations and identify where the top is?
- Convert the following infix into a postfix expression. $\rightarrow B - (C/D + (E \% F * G) / H) * J$
- Evaluate the given postfix expression. $53+62/*35*+$
- Construct a B-Tree of Order 3 by inserting numbers 34,26,45,12,87,56,78,22,23,24,65,85,95.
- Construct a 2-3-4 Tree by inserting the following sequence of numbers 45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 48.
- Build a max heap for the following numbers - 20, 35, 23, 22, 4, 45, 21, 5, 42 and 19. At each stage of heap tree construction, ensure heap properties are satisfied.
- Perform the operations onto splay tree (draw after each operation): insert (3), insert (9), insert (12), insert (55), insert (1), insert (3), delete (3), delete (9), delete (60), insert (3), search (12).
- Describe the algorithm to sort the elements 170, 45, 75, 90, 802, 24, 2, 66 using Radix sort. Explain with each phase?
- Consider the following graph & construct BFS.



Graph with six nodes

Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30	Demonstrates complete and accurate understanding of concepts	Good understanding with minor gaps	Basic understanding	Limited understanding
Accuracy of Answer	25	Answers are completely correct and relevant	Mostly correct with minor errors	Partially correct	Mostly incorrect
Problem-Solving & Reasoning	20	Provides logical reasoning and appropriate steps	Good reasoning with minor gaps	Basic reasoning	Lacks logical reasoning
Clarity & Relevance	15	Answers are precise, clear, concise, and directly address the question	Mostly clear and relevant	Somewhat unclear or incomplete	Irrelevant or unclear answers
Time Management & CompletionS	10	Completes all questions accurately within the time	Completes most questions on time	Requires additional time	Unable to complete the test

SFaculty In-charge	Course Coordinator	Program Director
Signature:	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

perform the following operations in the Singly linked list for the following input.

input : $22 \rightarrow 77 \rightarrow 55 \rightarrow 88 \rightarrow$

i) Insert a new-node 110 at the end of the list.
Given,

$22 \rightarrow 77 \rightarrow 55 \rightarrow 88 \rightarrow \text{Null}$

$22 \rightarrow 77 \rightarrow 55 \rightarrow 88 \rightarrow 110 \rightarrow \text{NULL}$

ii) Insert 101 at Position 4.

Position 4 means insert before the existing 4th node (88).

$22 \rightarrow 77 \rightarrow 55 \rightarrow 101 \rightarrow 88 \rightarrow 110 \rightarrow \text{NULL}$

iii) Delete node at the beginning.

Delete 22.

$77 \rightarrow 55 \rightarrow 101 \rightarrow 88 \rightarrow 110 \rightarrow \text{NULL}$

iv) Delete node at Position 3

Position 3 is 101.

$77 \rightarrow 55 \rightarrow 88 \rightarrow 110 \rightarrow \text{NULL}$

2. Rotations in Each step of insertion. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 ($\circ$), 4) Push (90°), 5) Push (36°), 6) Push (11°), 7) Push (88°), 8) Pop ($\circ$), 9) Pop ($\circ$). Draw the diagram of the stack and illustrate the above operations and identify where the top is?

Stack operations

Assuming the stack initially contains:

1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

where 10 is TOP.

Initial Stack

TOP → 10
9
8
7
6
5
4
3
2
1

Push(90)

TOP → 90
10
9
8
7
6
5
4
3
2
1

Push(36)

TOP → 36
90
10
9
8
7
6
5
4
3
2
1

Push(11)

TOP → 11
36
90
10
9
8
7
6
5
4
3
2
1

Push(88)

TOP → 88
11
36
90
10
9

8
7
6
5
4
3
2
1

Final stack

Top $\rightarrow$ 36

90

10

9

8

7

6

5

4

3

2

1

POP()

88 is removed.

POP()

11 is removed

TOP = 36

3) Convert the following infix into Postfix Expression $\rightarrow B - [C/D + [E \div F * G] / H] * J$

Given,

$$B - (C/D + (E \div F * G) / H) * J$$

* Inside the inner Parentheses:

$$E \div F * G$$

* Since $\div$ and $*$ have the same precedence, go left to right:

$$E F \div G *$$

Next:

$$C/D$$

becomes:

$$C D /$$

And:

$$(E \div F * G) / H$$

becomes:

$$E F \div G * H /$$

Now:

$$C/D + [E \cdot F \cdot G] / H$$

becomes:

$$C D / E F \cdot G \cdot H / +$$

Finally

$$B C D / E F \cdot G \cdot H / + J \cdot$$

4) Evaluate the given Postfix Expression

$$53+62/*35*+$$

Given: $53+62/*35*+$

Symbol	operation	stack
5	Push	5
3	Push	5 3
+	$5+3=8$	8
6	Push	8, 6
2	Push	8, 6, 2
/	$6/2=3$	8 3
*	$8 \times 3 = 24$	24
3	Push	24 3
5	Push	24 3 5
*	$3 \times 5 = 15$	24 15
+	$24+15=39$	39

Final answer

39

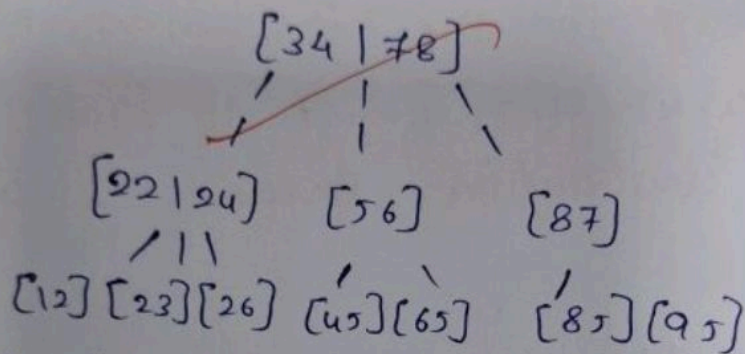
Construct a B-Tree of order 3 by inserting numbers 34, 26, 45, 12, 87, 56, 78, 22, 23, 24, 65, 85, 95.

Insert :

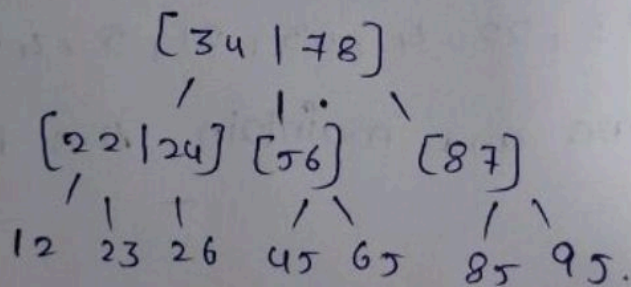
34, 26, 45, 12, 87, 56, 78, 22, 24, 65, 85, 95

Order-3-B Tree at most 2 keys.

B-Tree



Final structure

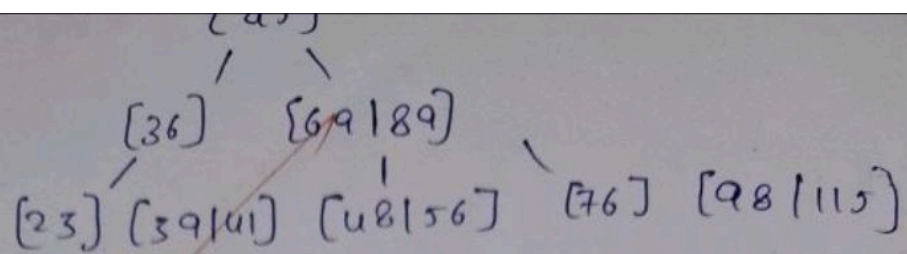


6. Construct a 2-3-4 Tree by inserting the following sequence of numbers 45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 48.

Insert :

45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 48

Final 2-3-4 Tree



all nodes contain at most 3 keys, satisfying the 2-3-4 Tree property.

7. Build a max heap for the following numbers - 20, 35, 23, 22, 4, 45, 21, 5, 42, 19. At each stage of heap tree construction, ensure heap properties.

Max heap Construction

20, 35, 23, 22, 4, 45, 21, 5, 42, 19

Insert one by one and maintain the max-Heap property.

Stages

20

35 20

35 20 23

35 20 23 20

35 20 23 20

35 22 23 20 4

45 22 35 20 4 23

45 22 35 20 4 23 21

15 22 35 20 4 23 21 5
45 42 35 ~~22~~ 4 23 21 5 20

45 42 35 22 19 23 21 5 20 4

8) Perform the operations onto splay tree (draw after each operation): insert (3), insert (9), insert (12), insert (55), insert (1), delete (9), search (12).

Splay tree operations

Insert (3)

Insert (9)

Insert (12)

Insert (55)

Insert (1)

~~Insert (3)~~

~~Insert (3)~~

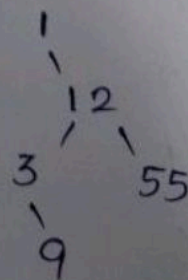
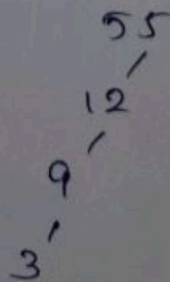
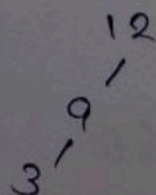
Delete (9)

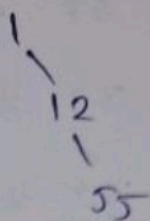
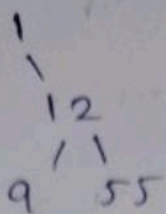
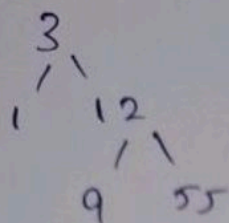
Delete (60)

Insert (3)

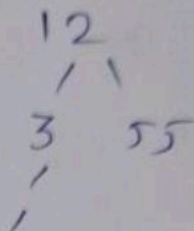
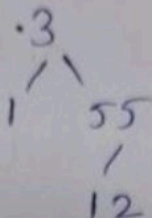
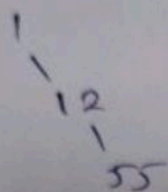
Search (12)

Insert (3)





60 does not exist



9 Describe the algorithm to sort the Elements 170, 45, 75, 90, 802, 24, 2, 66 using Radix Sort. Explain with Each Phase?

Given,

170, 45, 75, 90, 802, 24, 2, 66

Radix sort Processes digits from least Significant digit (LSD).

Phase 1 - units digit

170 $\rightarrow$ 0

45 $\rightarrow$ 5

75 $\rightarrow$ 5

90 $\rightarrow$ 0

802 $\rightarrow$ 2

24 $\rightarrow$ 4

2 $\rightarrow$ 2

66 $\rightarrow$ 6

After sorting

170, 90, 802, 2, 24, 45, 75, 66

Phase 2 - Tens digit

- 170 $\rightarrow$ 7
- 90 $\rightarrow$ 9
- 802 $\rightarrow$ 0
- 2 $\rightarrow$ 0
- 24 $\rightarrow$ 2
- 45 $\rightarrow$ 4
- 75 $\rightarrow$ 7
- 66 $\rightarrow$ 6

Phase 3 - Hundreds digit

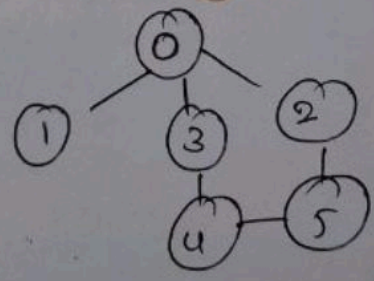
- 802 $\rightarrow$ 8
- 2 $\rightarrow$ 0
- 24 $\rightarrow$ 0
- 45 $\rightarrow$ 0
- 66 $\rightarrow$ 0
- 170 $\rightarrow$ 1
- 75 $\rightarrow$ 0
- 90 $\rightarrow$ 0

802 $\rightarrow$ 2, 24, 45, 66, 170, 75, 90

Final sorted order:

2 $\rightarrow$ 24 $\rightarrow$ 45 $\rightarrow$ 66 $\rightarrow$ 75 $\rightarrow$ 90 $\rightarrow$ 170 $\rightarrow$ 802

10. Consider the following Graph & Construct BFS.



BFS Traversal

Step 1:- Start at 0

Queue: [0]

Step 2:- Visit 0 $\rightarrow$ adjacent nodes are 1, 3, 2

Queue: [1, 3, 2]

Step 3:- Visit 1 $\rightarrow$ no new node.

Queue: [3, 2]

Step 4:- Visit 3 $\rightarrow$ new node 4

Queue : [2, 4]

Step 5:- Visit 4 $\rightarrow$ new node 5 already visited

Queue : [5]

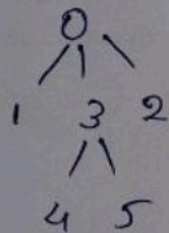
Step 6:- Visit 5 $\rightarrow$ all adjacent nodes visited

Queue : []

Final BFS order

0 $\rightarrow$ 1 $\rightarrow$ 3 $\rightarrow$ 2 $\rightarrow$ 4 $\rightarrow$ 5

BFS Tree:-



Wojas